

Fraud detection system

Tech stack

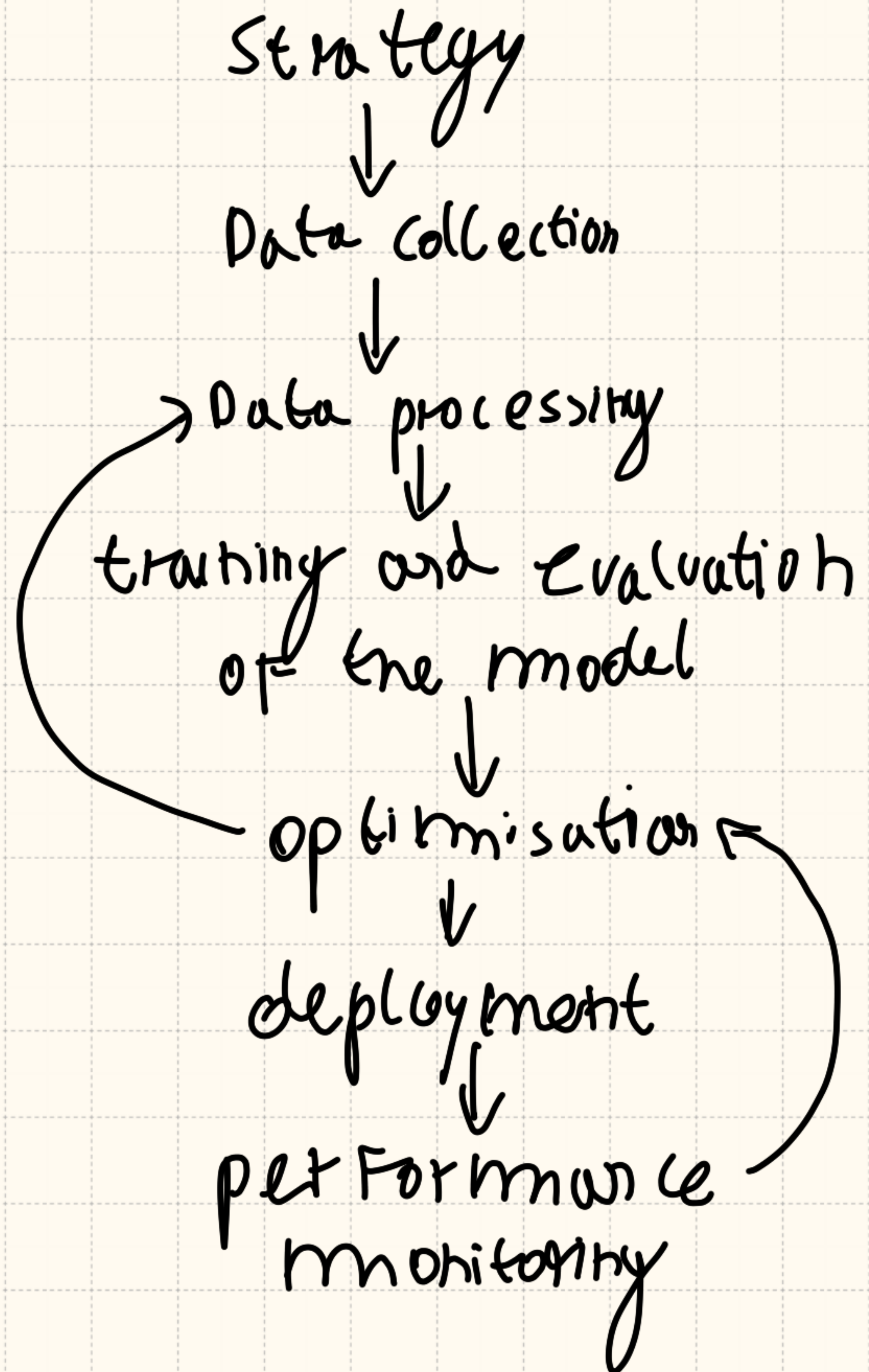
pandas

numpy

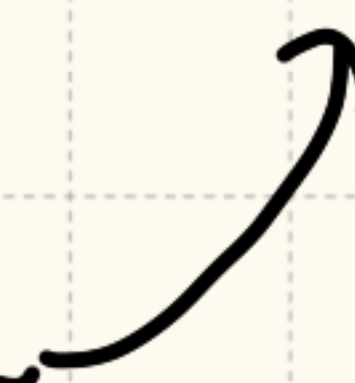
Scikit-learn

matplotlib

Joblib



Dataset → From kaggle

1. | install python-dotenv
 2. paste kaggle credentials → create .gitignore and include
 3. Run python Import code the file
 4. load downloaded data
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Pandas Phase 1: Ingestion & Exploratory data analysis PHASE 1

Series: 1D array holding any datatype

1. $a = [1, 2, 7]$ ^{input} | 2. `my series = pd.Series(a)` <sup>Index [...]
to rename
index name</sup>

3. `print(my series)` | output:

0	1
1	2
2	7

 Matrix
creation

Data Frames: 2D data structure with rows & col

Input: any csv data | `df = pd.DataFrame(data)`

Index	label 1	label 2
0	data 1	
1	.. 2	
2	.. 3	

Locate rows: `print(df.loc[0])`
able to locate it by index name

Read CSV files: `df = pd.read_csv('data.csv')`

PHASE 1

exploratory data analysis

step 0:

• identified table structure

Key file transaction format

	transaction-id	customer-id	card number	...
0				
1	"	"	"	
2	"	"	"	
3	"	"	"	
4	"	"	"	

step 1:

• checking for null, incomplete or invalid values.

⚠ to prevent garbage in - garbage out

5 rows x 24 columns → warning set

Pre-processing

PHASE 2

- checking total number of True/False
 - TO see distribution over whole dataset

High cardinality identifiers

- identify the columns that carry no learning pattern and drop them
 - columns that are unique to an individual and not the scope

Drop: trans-id, cust-id, card-num, device-fingerprint, IP-address
all these columns provide 0 info

Categorical encoding

With the remaining columns convert the categories into numbers

Do not change original DataFrame
Keep it for comparison

One hot encoding: Nominal/unordered data

↑ Best Fit For Binary data

- Turns one col into two separate col
eg. (is-Fraud-True / is-Fraud-False)

- No need to convert bool flags

distance-from-home

high-risk-merchant, ~~transaction-hour~~

weekend-transaction, ~~is-fraud~~

Character: contains string label that can be converted into 1/0

numerical Feature PHASE 2
hot categorical label

values from 0-23

16 columns

↳ no need to encode target variable

Fitting Phase: Analyse training data to learn parameters

- No data change, only stores

Fit - transform(df)

Transformation Phase: applied learned rules to alter dataset

never on test data.

- convert raw data into clean scaled format

Unpacking json data

> standard json wrapped

in double quotes

If you find yourself with singles
use ast.literal_eval to extract json

→ Both do same job

json.Cood

standard json

ast.literal_eval

single quote wrapped json

Train-Test Split & Feature preparation

1. Isolating target column: IS-Fraud
From dataset

2. Stratified Train-Test split
Split data: 80% training
20% test

1000 rows
80%
800 training rows

20%
200 testing rows

target Feature (y)
separate Feature (x)

Stratification:
Maintained exact
Fraud-to-Legitimate

Classification Model For Fraud Detection

Logistic Regression:

Random Forest

Gradient Boosting

After running evaluation block: output

Precision, Recall, F1-score
across the 3 algos

Precision

Recall

F1

1 Logistic Regression

0.80

0.92

0.86

1 Random Forest

0.94 ↑

0.85

0.89

1 Gradient Boosting

0.89

0.94 ↑

0.92 ↑